THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS CHAPTER TEST SET A

Chapter 3: **Number Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A number that reads the same forwards and backwards is a: 1

(a) triangular number **(b)** prime
(c) supercell **(d)** palindrome

2. The Collatz conjecture is still: 1

(a) an unsolved problem **(b)** proved long ago
(c) false **(d)** a school test

3. The largest 4-digit number from 3, 8, 1, 5 is: 1

(a) 8351 **(b)** 5381 **(c)** 8531 **(d)** 8513

4. Rounding 3482 to the nearest hundred gives: 1

(a) 3480 **(b)** 3500 **(c)** 3000 **(d)** 3400

5. Which of the following is an even number? 1

(a) 5138 **(b)** 5137 **(c)** 5139 **(d)** 5135

6. ‘Mental math’ means: 1

(a) guessing
(b) calculating quickly in the mind using tricks
(c) only using a calculator
(d) using an abacus

7. Which is the largest 5-digit number? 1

(a) 10000 **(b)** 99999 **(c)** 99990 **(d)** 98765

8. The sum of two consecutive counting numbers is always: 1

(a) odd **(b)** a multiple of 3 **(c)** a multiple of 4 **(d)** even

9. The Collatz sequence starting at 10 takes how many steps to reach 1? 1

(a) 9 **(b)** 5 **(c)** 7 **(d)** 6

10. One step of the Kaprekar routine on 3524 gives: 1

(a) 3096 **(b)** 3087 **(c)** 3177 **(d)** 2997

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a palindrome? Give two examples. 2

12. Write the largest and smallest 4-digit numbers using the digits 2, 6, 9, 4. 2

13. Apply one step of the Kaprekar routine to 4153. 2

14. Write the Collatz sequence starting from 5. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Perform the Kaprekar routine on 3087 and describe what happens. 3

16. Write the Collatz sequence for 7 and count the steps. 3

17. Make 5 different 4-digit palindromes and find their sums with their reverses. 3

OR

Estimate 487 × 52 and compare with the exact value. 3

18. Explain with an example how a palindrome is formed by adding a number to its reverse repeatedly. 3

OR

Why does the sum of three consecutive numbers always divide by 3? Show with 14, 15, 16. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Explain the Kaprekar routine. 5
(ii) Apply it to 2,531.
(iii) Show that it reaches 6174.

OR

(i) Write the Collatz rules. 5
(ii) Find the sequence for 11.
(iii) Count the steps.

20. (i) What are palindromic numbers? 5
(ii) List all 3-digit palindromes with digit sum 8.
(iii) Find the sum of 247 and its reverse.

OR

Case study: A school bell rings every 15 minutes and a club bell every 20 minutes starting at 9:00. 5
(i) After how many minutes do they ring together?
(ii) At what time?
(iii) How many times between 9:00 and 12:00?
(iv) Which number idea is used?

END OF QUESTION PAPER — Total: 40 Marks

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